Analysis of Covariance

Combining ANOVA with regression: covariates, the common-slope model, adjusted means and the adjusted F test.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 What is ANCOVA?

ANCOVA combines **analysis of variance** with **regression adjustment** on a **covariate**: a measured explanatory variable.

Agri examples: initial plant stand, initial weight of animals, pre-treatment pest count, previous-season yield.

A covariate must be **related to the response** and **not caused by the treatment**. Ideally measured **before** treatment (pre-treatment covariate).

2 Blocking vs ANCOVA

	Blocking	ANCOVA
When	Before the experiment (layout)	During analysis
Handles	Predictable variation (soil, gradient)	Variation in a measured covariate
Aim	Both reduce experimental error	

Precision may improve but is **not guaranteed**.

3 Model & key quantities (CRD)

Common-slope model

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}$$

Slope

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

Adjusted error SS

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

Adjusted MSE

$$MSE_{adj} = \frac{E}{n - v - 1}$$

E_{xx}, E_{xy}, E_{yy} = within-treatment (error) sums of squares and products. RBD error df: $(r-1)(v-1) - 1$.

4 Degrees of freedom

Fitting one covariate slope **uses one error df**.

CRD	$n - v - 1$
RBD	$(r-1)(v-1) - 1$

5 Adjusted means & F test

Adjusted treatment mean

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

Adjusted F

$$F = \frac{(E_1 - E)/(v-1)}{MSE_{adj}}$$

E_1: residual SS of the reduced model (no treatments, covariate kept); E: full model. Means are compared at a **common covariate value**.

6 Assumptions

- ✓ The usual ANOVA assumptions: independent errors, common variance, normality for exact tests
- ✓ **Linear** relationship between response and covariate
- ✓ **Equal slopes** across treatments (check treatment × covariate interaction)
- ✓ Covariate not affected by the treatments
- ✓ Error mean is zero

7 Steps of the analysis

1. Compute X, Y and XY SS for
2. Get error E_{xx}, E_{xy}, E_{yy} SP for
3. Estimate $\hat{\beta} = E_{xy}/E_{xx}$
4. Adjusted MSE_{adj} error SS and
5. Adjusted treatment SS: $E_1 - E$
6. Adjusted F test
7. Compute adjusted means
8. Compare with pair-specific SEs

! Common mistakes

- × Using a covariate the treatment affects.
- × Forgetting the extra df lost.
- × Assuming adjustment turns observational data into an experiment.

? Fill in the blanks

1. Initial plant stand counted before applying treatments is a suitable _____. [covariate]
2. With E_xy = 30 and E_xx = 60, the slope b = _____. [0.5]
3. Treatment mean 50, its covariate mean 12, overall covariate mean 10 and b = 0.5 give an adjusted mean of _____. [49]
4. A CRD with 20 observations, 4 treatments and one covariate has _____ error df. [15]
5. A covariate must not be _____ by the treatments. [affected]
6. The common-slope model assumes the treatment regression lines are _____. [parallel]

Source: *Textbook of Agricultural Statistics*, Ch. 22 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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